

Index

A

- abductor hallucis strains, 956
- accelerating tissue healing, 40
- acceleration/deceleration injury (whiplash), 375
- accessory soleus, 888
- acetabular labrum, 594-5
- Achilles pain, 865-92
 - accessory soleus, 888
 - acute Achilles tendon rupture (complete), 889-92
 - autologous blood and platelet rich plasma (PRP), 882
 - causes, 867
 - cell-based therapies, 883
 - clinical assessment, 878-9, 886
 - clinical perspective, 865-72
 - compression, 884-6
 - corticosteroid injections, 882
 - electrophysical agents, 884
 - gradual onset, 888
 - history, 866-8
 - insertional Achilles tendinopathy, 884-7
 - investigations, 872
 - medications, 884
 - midportion Achilles tendinopathy, 872-84
 - pathology, 872-6
 - physical examination, 868-72
 - posterior impingement syndrome, 887-8
 - PROMs, 872
 - referred pain, 888
 - rehabilitation exercise training, 888
 - rehabilitation of Achilles tendon ruptures, 890-1
 - retrocalcaneal bursitis (enthesis organ), 884-5, 886-7
 - return to sport, 891-2
 - sclerosing, 882-3
 - Sever's disease, 888
 - surgical treatment, 884
 - tendinopathy, 872-84
 - treatment, 886
- Achilles (superficial calcaneal) bursitis, 887
- acquired sport-specific instability (AIOS), 407, 408, 411, 412, 418
- acromioclavicular joint injuries, 426-8
- active movement tests (neck pain), 355-7
- active recovery, 189-91, 200
- high-duration short-duration exercise, 190
- long-duration exercise, 190-1
- psychological effects, 191
- active spondylolytic bone stress injury, 541-2
- activity history and injury, 213
- actovegin, 269
- acupuncture, 256-8
 - lateral elbow tendinopathy, 447
- acute Achilles tendon rupture (complete), 889-92
- acute acromioclavicular joint injuries, 426
- acute ankle injuries, 893-915
 - anteroinferior tibiofibular ligament injury/syndesmotic injury, 912-13
 - anterior calcaneal process fracture, 910
 - avulsion fracture of the base of the fifth metatarsal, 908-10
 - causes, 906, 908, 912-14
 - clinical approach, 906
 - clinical perspective, 894-8
 - common/less common, 895
 - complex regional pain syndrome type 1, 914-15
 - examination, 895-7
 - fractures, 910-11
 - functional anatomy, 893-4
 - grade III injuries (treatment), 903-4
 - history, 895
 - impingement syndromes, 911
 - investigations, 897-8
 - lateral ligament injuries, 899-900
 - lateral talar process fracture, 910
 - maisonneuve fracture, 905
 - medial (deltoid) ligament injuries, 904-5
 - medial tubercle fractures, 911
 - osteochondral lesions of the talar dome, 906-8
 - Ottawa ankle rules, 895, 897, 898, 905
 - peroneal tendons dislocation, 911-12
 - persistent pain after ankle sprain, 905-15
 - posterior process of the talus fracture, 911
 - post-traumatic synovitis, 913-14
 - Pott's fracture, 905
 - rehabilitation, 900-4
 - sinus tarsi syndrome, 914
 - tendon dislocation/rupture, 911-12
 - tibial plafond lesions, 910
 - tibialis posterior tendon dislocation, 912
 - tibialis posterior tendon rupture, 912
 - treatment, 900-4, 907-908
 - see also* ankle pain; ankle sprain
- acute compartment syndrome, 24-5, 844
- acute elbow injuries, 456-60
 - fractures, 456-7
 - hyperextension injuries, 460
 - rupture of MLC, 459
 - posterior dislocation, 457-9
 - tendon ruptures, 459-60
- acute groin injuries, 633, 642-3
 - diagnosis, 642-3
- acute hamstring muscle strains, 685-705
 - criterion-related rehabilitation, 693
 - epidemiology, 686
 - manual therapy, 694-5
 - medical therapies, 692-3
 - muscle activation, 692
 - prognosis, 689-703
 - RICE, 692
 - risk factors, 703-5
 - stretching, 694
 - type I and type II, 686-8
 - type I: sprinting related, 688
 - type II: stretch-related (dancer's), 689
- acute knee injuries, 713-67
 - acute patellar trauma, 762-5
 - anterior cruciate ligament tears, 737-55
 - articular cartilage damage, 760-2
 - bursal haematoma, 767
 - causes, 714
 - clinical approach, 714-23
 - fat pad impingement, 767
 - fracture of the tibial plateau, 767
 - functional anatomy, 713-14
 - investigations, 722-3
 - lateral collateral ligament tears, 759-60
 - medial collateral ligament injury, 729-36
 - meniscal injuries, 723-8
 - patellar tendon rupture, 765, 767
 - physical examination, 716-22
 - posterior cruciate ligament (PLC) tears, 755-9

- prevention, 177-81
- rehabilitation, 727-88, 745-50
- ruptured hamstring tendon, 767
- superior tibiofibular joint injury, 767
- surgical treatment, 742-4
- time relationship of swelling to
 - diagnosis, 715
- treatment, 726-7, 729-36, 757-9, 765
- acute patellar tendon pain, 801
- acute patellar trauma, 762-5
- acute radiculopathy +/-nerve root
 - compression, 545-7
- acute sports injuries *see* sports
 - injuries, acute
- acute wry neck, 374
- adductor magnus strains, 711
- adductor-related groin pain, 643-50
 - acute/subacute phase, 647
 - compression shorts, 650
 - conditioning phase, 647-8
 - diagnostic criteria, 643-50
 - manual therapy, 650
 - rehabilitation, 649
 - return-to-sport phase, 649
 - sports-specific phase, 648-9
 - treatment, 644-5
- adhesive capsulitis 'frozen
 - shoulder', 429
- Advanced Trauma Life Support
 - (ATLS) 299
- age and ACL injuries, 755
- agility training, 144
- amnesia and injury severity, 305-6
- analgesics, 264
- anatomical considerations (neck
 - pain), 347-9
- anatomical planes of the body, 86
- ankle dorsiflexion, 98-9
- ankle injuries *see* acute ankle injuries
- ankle joint, 85-8
- ankle pain, 917-36
 - anterior, 934-6
 - lateral, 929-33
 - medial, 917-29
 - posterior, 917
 - see also* Achilles pain; acute ankle
 - injuries
- ankle sprain, persistent pain after 'the
 - difficult ankle', 905-15
- avulsion fracture of the base of the
 - fifth metatarsal, 908-10
- causes, 912-14
- clinical approach, 906
- complex regional pain syndrome
 - type 1, 914-15
- fractures, 910-11
- impingement syndromes, 911
- osteochondral lesions of the talar
 - dome, 906-8
- post-traumatic synovitis, 913-14
- sinus tarsi syndrome, 914
- tendon dislocation or rupture,
 - 911-12
- ankle sprain prevention, 174-7
 - injury mechanisms, 175
 - programs, 175-7
 - risk factors, 175
- anterior ankle pain, 934-6
 - anterior impingement of the
 - ankle, 934-5
 - anterior inferior tibiofibular
 - ligament (AITFL) injury, 936
 - tibialis anterior tendinopathy, 935-6
- anterior calcaneal process fracture, 910
- anterior cruciate ligament (ACL)
 - injuries, 166, 714, 715
 - age, 755
 - anatomy, 737
 - clinical features, 738-40
 - gender difference, 754-5
 - mechanism of injury, 737-8
 - muscle weakness, 749-50
 - osteoarthritis, 753-4
 - partial tear, 755
 - prevention, 755
 - re-injury rate, 752-3
 - return to sport, 751-2
 - revision surgery, 753
 - self-reported knee function, 750-1
 - surgery, 748-50
 - tears, 737-55
 - treatment, 740-4
 - treatment, outcomes after, 750-5
- anterior cruciate ligament
 - rehabilitation, 745-5
 - instability, 748
 - low back pain, 748
 - low limb stiffness, 749
 - muscle weakness, 749-50
 - patellar region pain, 748-9
 - postural control, 750
 - psychological problems, 750
 - soft tissue laxity, 749
 - soft tissue stiffness, 749
 - symptomatic meniscus tears, 748
- anterior exertional compartment
 - syndrome, 843-4
- anterior impingement of the ankle,
 - 934-5
- anterior inferior tibiofibular ligament
 - (AITFL) injury/syndesmotic
 - injury, 912-13, 936
- anterior knee pain, 769-803
 - bursitis, 802-3
 - causes, 770
 - clinical approach, 770-5
 - fat pad irritation/impingement,
 - 801-802
 - Osgood-Schlatter lesion, 802
 - patellofemoral instability, 792-3
- patellofemoral pain, 775-91
- patient reported outcome
 - measures, 774-5
- physical examination, 771-4
- quadriceps tendinopathy, 802
- Sinding-Larsen-Johansson lesion, 802
- synovial plica, 803
- anterior thigh pain, 659-77
 - biomechanics, 660-1
 - causes, 660
 - clinical approach, 661-4
 - epidemiology, 660
 - femoral nerve injury, 676-7
 - functional anatomy, 660-1
 - investigations, 664
 - key outcome measures, 664
 - lateral femoral cutaneous
 - nerve injury ('meralgia
 - paresthetica'), 676
 - physical examination, 661-4
 - quadriceps contusion, 664-71
 - quadriceps strain, 671-4
 - referred apin, 677
 - stress fracture of the femur, 674-6
- anterolateral impingement, 932-3
- antibiotics, quinolone, 272
- anti-doping, 1020
- antidepressants, 267
- anti-gravity treadmill training, 41
- apophysitis, 982-3
 - of the fifth metatarsal, 989
 - of the tarsal navicular bone, 989
- arm pain, 439-62
 - forearm, 460-1
 - upper, 461-2
- arthroscopic surgery, 274-5
- arthroscopy
 - diagnosis (knee), 723
 - diagnosis (shoulder), 402
 - hip injury, 621-2
- artificial turf *vs.* natural grass, 187-8
- articular cartilage
 - damage, 760-2
 - and mechanotherapy, 245
- atherosclerotic vessel disease, 861-2
- athlete
 - coach and the clinician
 - (relationship), 5-6
 - and importance of sport
 - (injuries), 213
 - overhead athletes and shoulder
 - injuries, 433-8
 - return to play, 285-93
 - and return-to-play decisions, 292
- athletes, periodic medical assessment
 - (PMA), 1003-15
 - action points, 1014-15
 - age, 1012
 - assessment of known injuries and
 - illnesses, 1006

available resources, 1012
 baseline testing, 1007-8
 cardiac screening, 1009
 clearance or restriction from
 play, 1013
 conditions affecting sports
 participation, 1003-6
 consent, 1013
 education, 1007
 geographical location, 1011-12
 identification of medical
 conditions, 1003-6
 insurance medical assessment, 1014
 pre-employment medical
 assessment, 1014
 rapport development, 1008
 review of current medications and
 supplements, 1007
 risk factors for future
 injury, 1009-11
 screening, 1008-11
 sex, 1012
 sport and position, 1011
 when to perform a PMA?, 1012-13
 who is being assessed, 1011-12
 athletes and core stability, 164
 atraumatic multidirectional instability
 (AMBRI), 411
 auricular haematoma, 335
 autologous blood and platelet rich
 plasma (PRP), 882
 autologous blood injections, 273, 450
 axillary nerve injury, 430-1
 axillary vein thrombosis ('effort'
 thrombosis), 432
 axio-scapular muscles (neck pain),
 361-2, 372

B

back pain, 981-2
 back pain, low, 521-65
 acute radiculopathy +/-nerve root
 compression, 545-7
 acute severe, 549-51
 clinical approach, 528-35, 563-5
 contributing factors, 526-8
 epidemiology, 521-3
 high-complexity profile, 562-3
 investigations, 534-5
 management of non-specific
 disorders, 548-63
 management of specific disorders,
 535-48
 medium-complexity profile, 561-2
 multidimensional nature, 523
 passive treatments, 551-4
 patient-reported outcome measures
 (PROMs), 529-30
 physical examination, 530-4

recurrent/persistent, 555-61
 rehabilitation, 563, 748
 spondylolisthesis, 543-4
 stress fracture of the pars
 interarticularis, 535-43
 subacute, 551
 triage, 523-6
 vertebral endplate oedema, 547-8
 young athlete, 981-2
 Bahr, Roald, 1034
 Baker's cyst, 822-3
 Bankart lesion, 401, 407, 408,
 409-10, 415
 Bankart repair, 408
 Baxter's nerve entrapment, 947-8
 biceps femoris tendinopathy,
 815, 823
 biceps rupture, 429
 bifurcate ligament injuries, 952
 biomechanical aspects of injury in
 specific sports, 121-38, 166
 cricket fast bowling, 125-7
 cycling, 121-4
 golf, 128-30
 rowing, 130-3
 swimming, 133-6
 tennis, 136-8
 biomechanical examination,
 214, 221
 biomechanics
 altering, 245-6
 bone stress injuries, 32
 and leg pain, 826
 biomechanics, clinical, 85-120
 footwear assessment, 105-6
 lower limb, 85-90
 lower limb abnormalities, 108-14
 lower limb assessment, 95-105
 with movement, 90-5
 upper limb, 114-20
 BioMed Central (BMC), 11
 bisphosphonates, 272
 blisters, 52
 blood flow and massage, 191
 bone injuries, 14, 15-17
 fracture, 14, 15-17
 and mechanotherapy, 245
 and NSAID, 266
 bone scan, 35, 642
 bone stress injury, 29-43
 apophysitis, 43
 articular cartilage, 43
 classification, 37-8
 diagnosis, 34-7
 epidemiology, 31
 examination, 35
 healing of, 42
 imaging, 35-7, 38
 osteitis and periostitis, 42-3
 pathophysiology, 30-1

bone stress injuries
 (management), 38-42
 accelerating tissue healing, 40
 healing, 42
 initial, 39-40
 and return to running, 40-2
 bone stress injuries (risk factors), 31-4
 biomechanical factors, 32
 calcium and vitamin D status, 34
 energy availability, 34
 load applied to bone, 32-3
 muscle factors, 33
 physical activity history, 34
 playing surfaces, 33
 resist load without damage
 accumulation, 33-4
 shoes and inserts, 33
 specific treatment, 38
 training factors, 32-3
 bone tumours, 78-80
 botulinum toxin (botox) injection, 450
 Boutonnière deformity, 503-4
 brachial plexus injuries (stingers/
 burners), 375
 bracing
 lateral elbow tendinopathy, 447-8
 patellofemoral pain, 788-9
 'stinger' phenomenon (neck pain), 375
 brain
 and pain, 59-61
 and persistent pain, 60-1
 and spinal cord, 60
 bruise/contusion, fat pad, 26-7
 'burners' phenomenon (neck pain), 375
 bursa, 14, 26, 51-2
 bursal hematoma, 767
 bursitis, 802-3, 819
 buttock pain, 567-91
 avulsion fracture of ischial
 tuberosity, 589
 causes, 567
 clinical approach, 567-74
 history, 568-9
 investigations, 573-4
 ischiofemoral impingement, 587
 myofascial, 574-6
 physical examination, 569-73
 piriformis syndrome, 585-7
 posterior thigh compartment
 syndrome, 587-8
 proximal hamstring
 tendinopathy, 577-81
 proximal hamstring tendon
 rupture, 588-9
 referred pain from lumbar spine, 576
 sacral and pubic ramus stress
 fractures, 590
 sacroiliac joint dysfunction, 581-5
 specificity and sensitivity of tests, 574
 spondyloarthropathies, 589-90

- C**
- calcaneal stress fracture, 947
 - calcium, bone stress injuries, 34
 - calf pain, 847-63
 - anatomy, 847-9
 - causes, 850, 862-3
 - clinical approach, 849-53
 - deep venous thrombosis, 863
 - gastrocnemius muscle strains, 853-6
 - investigations, 853
 - nerve entrapments, 862-3
 - neuromyofascial causes, 862
 - physical examination, 850-3
 - soleus muscle strains, 856-9
 - superficial compartment syndrome, 863
 - career development, 1027-34
 - key behaviours for a successful and interesting career, 1029-31
 - lessons from around the world, 1031-4
 - sport and exercise medicine, 1027-9
 - carpal bones, dislocation, 479
 - carpal tunnel syndrome, 471, 485-6
 - carpometacarpal joint dislocations, 498-9
 - case manager approach and SEM, 6-8
 - cell therapy, 274
 - Central Government Model (CGM) for the limits of performance 140-2
 - central nervous system (CNS), 70
 - central sensitization, 69-71
 - centrally dominated pain, 69-72
 - cervical arterial dysfunction, 364
 - cervical extensors, assessment of, 360-1
 - cervical flexors, assessment of, 358-60
 - cervical nerve injury, 374-5
 - cervical radiculopathy, 374-5
 - cervical sensorimotor function, tests for, 362-4
 - cervicogenic headache, 322-8
 - classification, 323
 - clinical features, 323-8
 - examination, 324-5
 - mechanism, 322-3
 - and neck pain, 374
 - treatment, 326-8
 - chest pain, 514-20
 - cardiac origin, 515
 - causes, 515
 - clinical assessment, 516
 - costochondritis, 518
 - stress fracture to ribs, 518-19
 - rib trauma, 516-17
 - side strain, 519-20
 - sternoclavicular joint problems, 517-18
 - thoracic spine, 517
 - chest wall pain, 132
 - children and concussion in sport, 311-16
 - chondral injuries, 17-18, 818-19
 - chondral surfaces, 596
 - chondroitin, 272-3
 - chronic acromioclavicular joint pain, 426-8
 - chronic compartment syndrome (CCS), 45, 861
 - chronic exertional compartment syndrome (CECS), 834-6, 841-5
 - chronic temporomandibular injuries, 345-6
 - chronic traumatic encephalopathy (CTE), 310
 - claudicant-type calf pain, 859-62
 - atherosclerotic vessel disease, 861-2
 - endofibrosis, 862
 - popliteal artery entrapment, 859-61
 - vascular causes, 859-62
 - clavicle fractures, 424-6
 - clinical approach
 - acute Achilles tendon rupture, 889
 - acute knee injury 714-23
 - anterior knee pain, 770-4
 - anterior thigh pain, 661-4
 - difficult ankle, 906
 - headache, 318-20
 - hip-related pain, 598-604
 - lateral knee pain, 806-9
 - leg pain, 823-24
 - lower back pain, 528-35
 - posterior thigh pain, 681-5
 - wrist pain, 463-70
 - clinical assessment, 201-7
 - Achilles tendon 886
 - calculating an accurate diagnosis, 202-5
 - chest pain, 516
 - and concussion, 311
 - differential diagnosis, 201-2
 - differential diagnosis: three step process, 201-2
 - facial injuries, 331-2
 - formal diagnosis assessment, 206
 - neurological, 357
 - positive and negative predictive values, 202-3
 - posterior knee pain, 820-2
 - reliability, 202
 - clinical examination
 - diagnosis, 4-5
 - groin pain, 635-9
 - clinical features
 - lateral elbow tendinopathy, 444-5
 - lisfranc joint injuries, 955
 - navicular stress fracture, 953-4
 - plantar fasciopathy, 942-3
 - clinical measurement of headache, 319-20
 - clinical neurological examination, 357
 - clinical perspective
 - Achilles pain, 865-72
 - acute ankle injuries, 894-898
 - forefoot pain, 958-9
 - neck pain, 349-65
 - clinical utility, 203-4
 - clinical relevant pathology (lateral elbow tendinopathy), 443-4
 - clinicians
 - coach, and the athlete (relationship), 5-6
 - and recovery, 198-9
 - and return-to-play decisions, 290-1
 - closed chain exercises, 146-8
 - coach
 - athlete and the clinician (relationship), 5-6
 - and return-to-play decisions, 292
 - Cochrane Library, 10, 12
 - codeine, 264
 - cognitive function and concussion, 308
 - cognitive therapy and pain, 75
 - cold water immersion (CWI), 195-6, 200
 - Colour Doppler ultrasound, 228
 - compartment syndrome
 - acute, 24-5, 844
 - chronic, 45, 861
 - chronic exertional compartment syndrome (CECS), 834-6, 841-6
 - of the posterior thigh, 711
 - of the thigh, 670
 - superficial compartment syndrome, 863
 - complex regional pain syndrome type 1, 914-15
 - compression
 - acute injury management, 248-50, 692
 - retrocalcaneal bursitis, 884-6
 - compression and recovery, 196, 200
 - compression shorts, 650
 - computed tomography (CT), 35, 37, 129, 221, 223, 229, 468-9, 642
 - concussion, 296-316
 - acute, 306
 - amnesia and injury severity, 305-6
 - Child-SCAT3™, 312-15
 - and children, 311-16
 - chronic traumatic encephalopathy (CTE), 310
 - clinical assessment, 311
 - definition, 296-7
 - diagnosis conformation, 300-6
 - initial impact, 298
 - investigations, 311
 - management of the athlete, 298-310

mental health issues, 310
 on-field management, 298-300
 prevention, 297-8
 prolongation of symptoms, 310
 return to play, 306-10
 SCAT3™, 301-4, 311
 second impact syndrome (SIS), 310
 symptoms and signs, 300, 308
 treatment, 311
 concussive convulsions, 310
 conditioning training, 142-50
 agility, 144
 endurance, 142-3
 flexibility, 149-50
 resistance, 144-9
 speed, 143-4
 contrast water therapy (CWT),
 195-6, 200
 contusion, muscle, 23-4
 core stability, 153-64
 in athletes, 164
 introduction, 153-4
 motor control training, 154-64
 prevention of pain and injury, 164
 corneal injuries, 338
 corns and calluses, 970
 coronoid fractures, 457
 corticosteroid injection
 lateral elbow tendinopathy, 448-9
 midportion Achilles
 tendinopathy, 882
 corticosteroid, 270-1
 costochondritis, 518
 costovertebral and costotransverse
 joint disorders, 511-12
 counter irritants, 267-8
 cramps, muscle, 25, 849
 cranio-cervical flexion test
 (CCFT), 358-9
 cricket fast bowling (biomechanical
 aspects of injuries), 125-7
 cross-training, 143
 cryotherapy modalities used to treat
 sports injuries, 249, 799
 cuboid, stress fracture of the, 956
 cuboid syndrome, 956
 cuneiforms, stress fracture of the, 956
 cupping, vacuum, 255
 cycling (biomechanical aspects of
 injuries), 121-4
 bike fitting, 124
 knee pain, 121-3
 low back pain, 123
 risk factors and loading, 121

D

de Quervain's tenosynovitis 476-7
 deep posterior compartment
 syndrome, 842-3

deep venous thrombosis, 824, 863
 delayed onset muscle soreness
 (DOMS), 46, 146, 189, 191-2
 dermatitis, 52
 diagnosis, 209-29
 accurate, 210-11
 clinical questions, 12
 description of symptoms, 211-12
 examination, 4-5, 213-21
 history, 211-13
 radiological investigation, 223-9
 sport and exercise medicine
 (SEM), 4-5
 diagnostic assessment
 clinical, 4-5, 201-7
 formal, 206
 diagnostic imaging, 221-3
 diagnostic metrics, 206
 differential diagnosis, 201-2
 three step process, 201-2
 digital ischaemic pressure, 253-4
 dislocation/subluxation 14, 18-19,
 457-9
 distal quadriceps muscle strain,
 671-2
 distal radioulnar joint instability, 484
 distal radius fracture, 470-2
 dorsal calcaneocuboid ligament
 injury, 952
 DRABC (danger, response, airway,
 breathing, circulation) 299
 dry needling, 256-8
 Dynamic Gait Index 364
 dynamic movement assessment, 104-5
 dynamic stretching, 150

E

ear injuries, 335-6
 elbow pain, 980
 elbow injuries, 439
 acute, 456-60
 elbow anatomy, 439-40
 elbow pain, 439-62
 lateral, 136-7, 440-51
 medial, 451-4
 posterior, 454-6
 electromagnetic therapies, 16, 263
 electrophysical agents, 261-4, 884
 electrotherapeutic modalities (lateral
 elbow tendinopathy), 446
 elevation (acute injury management),
 247, 250, 692
 emergency action plans and teams,
 1020-1
 Emergency Management of Severe
 Trauma (EMST) 299
 endocrine disorders, 81-2
 endofibrosis, 862
 endurance, skill training as, 143
 endurance training, 142-3
 methods, 142-3
 enthesopathy, 51
 epidemiology
 anterior thigh pain, 660
 bone stress injuries, 31
 groin pain, 631-2
 low back pain, 521-3
 equipment
 and injury, 213, 221
 injury prevention and protective,
 187
 evidence-based medicine (EBM), 9
 evidence-based practice (EBP), 9-12
 challenges to, 10
 concept of, 10
 definition, 9
 implementing, 10-12
 examination and injury, 213-21
 biomechanical, 214, 221
 examination routine, 214-21
 functional testing, 214
 groin pain, 635-9
 lateral hip pain, 623-5
 ligament testing, 215
 palpation, 215
 range of motion, 214-15
 spinal, 214, 219-21
 strength testing, 215-19
 excessive lateral pressure
 syndrome, 814-15
 exercise
 high-intensity short-duration, 190
 long-duration, 190-1
 exercise associated muscle cramps
 (EAMC), 25
 exercise-induced anaphylaxis
 (EIA), 213
 exercise-induced hypoalgesia, 246
 exercise-induced muscle
 soreness, 45-6
 exercise medicine, 1029
 exercises *see* resistance exercises
 exertional compartment
 syndrome surgery
 outcomes, 844-5
 rehabilitation, 845
 extensor carpi ulnaris tendon injuries,
 484-5
 extensor tendinopathy, 956
 external compression headache, 328
 extracorporeal shockwave therapy
 (ECSWT), 263-4
 eye injuries, 336-40
 assessment, 336-8
 corneal injuries, 338
 hyphema, 339
 lens dislocation, 339
 prevention, 340
 retinal detachment, 340

retinal haemorrhage, 339
 subconjunctival haemorrhage, 338-9
 vitreous haemorrhage, 339
 eyelid injuries, 339

F

faceguards, 341
 facial bones, fractures of, 343-6
 maxillary fractures, 344
 mandibular fractures, 344-5
 orbital fracture, 343-4
 temporomandibular injuries, 345-6
 zygomaticomaxillary complex, 344
 facial injuries, 331-46
 clinical assessment, 331-2
 ear, 335-6
 eye, 336-40
 functional anatomy, 331
 lacerations and contusions, 332-3
 nose, 333-5
 prevention, 346
 in sport, 333
 teeth, 340-3
 Fagan's nomogram, 203-4, 226
 fascia
 tear/rupture, 14, 26
 Fartlek training, 142-3
 fat pad, 26-7
 contusion (rear foot pain), 946-7
 impingement, 767
 irritation/impingement, 801-802
 fatigue and running biomechanics, 95
 femoral nerve injury, 676-7
 femoroacetabular impingement (FAI),
 607-10
 fibrocartilage injuries, 14, 18
 fibula fracture, 845
 finger injuries, 489-505
 Boutonnière deformity, 503-4
 carpometacarpal joint dislocations,
 498-9
 chronic swan neck
 deformity, 503
 clinical approach, 489-92
 dislocation, 499
 fractures of phalanges, 497-8
 'jersey finger', 504
 ligament and tendon injuries,
 499-504
 mallet finger, 501-3
 metacarpal fractures, 494-7
 overuse conditions, 504-5
 patient-reported outcome measures
 (PROMs), 493
 first branch of lateral plantar
 nerve (Baxter's nerve)
 entrapment, 947-8
 first tarsometatarsal joint pain, 955-6
 flat foot, 989

flexibility in the rehabilitation
 process, 150
 flexibility training, 149-50
 flexion-pattern motor control
 dysfunction, 123
 flexor hallucis longus
 tendinopathy, 924-6
 fluid replacement and
 recovery, 197
 foot
 biomechanical assessment, 96-8
 bones, 938
 first metatarsophalangeal joint
 (MTPJ) range, 98
 flat, 989
 foot orthoses, 109-12
 comfort, 112
 custom-fabricated, 109-10
 effectiveness, 110-11
 fitting (contemporary approaches),
 111-12
 mechanism of action, 110
 and over-use injuries, 183
 subtalar joint neutral, 111-12
 treatment direction test, 112
 types of, 109-10
 foot pain, 937-72
 forefoot, 958-72
 midfoot, 948-58
 rear, 937-48
 foot posture index, 96-8
 footwear as therapy, 112-14
 footwear assessment tool, 105-6
 forearm compartment pressure
 syndrome, 461
 forearm injuries and rowing, 132-3
 forearm pain, 460-1
 forefoot pain, 958-72
 causes, 958, 972
 clinical perspective, 958-9
 corns and calluses, 970
 first MTP joint sprain ("turf toe"),
 966-7
 fractures of the fifth metatarsal,
 961-5
 Frieberg's osteochondritis, 972
 hallux limitus, 967
 hallux valgus, 967-8
 Joplin's neuritis, 972
 Morton's interdigital
 neuroma, 970-1
 MTP joint synovitis, 965-6
 onychocryptosis 972
 plantar plate tear, 969-70
 plantar warts, 971
 sesamoid injuries, 968-9
 special metatarsal stress fracture:
 base of the second MT, 961
 stress fractures of the great
 toe, 972

stress fractures of the neck of
 metatarsals I-IV, 959-61
 subungual exostosis, 972
 subungual haematoma, 972
 toe clawing, 972
 fractures, 14, 15-16, 34
 acute, 976-9
 acute ankle injuries, 905, 907,
 910-11
 acute elbow injuries, 456-7
 anterior calcaneal process, 910
 apophyseal (avulsion), 978-9
 avulsion fracture of the base of the
 fifth metatarsal, 908-10
 avulsion fracture of ischial
 tuberosity, 589
 clavicle, 424-6
 coronoid, 457
 diaphyseal, 978
 distal radius, 470-2
 facial bones, 343-6
 fibula, 845
 hook of hamate, 481-2
 lateral talar process, 910
 maisonneuve, 905
 medial tubercle, 911
 metacarpal, 494-7
 metaphyseal, 976-7
 nasal, 334
 olecranon, 456-7
 patella, 762
 phalanges, 497-8
 physeal (growth plate), 977-8
 posterior process of the
 talus, 911
 Pott's, 905
 radial head, 457
 radius and ulna, 460
 scaphoid, 472-5
 shoulder, 425, 432
 supracondylar, 456
 talus, 933
 tibia, 846
 tibial plateau, 767
 trapezium, 475-6
 triquetral, 482
 ulnar styloid, 481
 see also stress fractures
 Freiberg's infarction, 989
 Freiberg's osteochondritis, 972
 'frozen shoulder', 429
 fuel replacement and
 recovery, 197-8
 function and injury, 212, 214
 functional anatomy
 acute ankle injuries, 893-4
 anterior thigh pain, 660-1
 bones of the face, 331
 hip-related pain, 593-8
 posterior thigh pain, 679-81

G

gabapentin, 267
 gait, painless abnormalities of, 986-8
 gait velocity and running, 94
 ganglions, 478-9
 gastrocnemius muscle strains, 853-6
 acute strain, 853-6
 operative intervention, 856
 physical examination, 853
 treatment, 853-6
 gastrocnemius tendinopathy, 824
 gender
 and ACL injuries, 755
 sport and exercise medicine, 1029
 and sports injuries, 166
 general health and injury, 212
 genetic disorders, 82
 glenohumeral internal rotation deficit (GIRD), 382
 clinical evaluation, 394-5
 pathological, 417-19
 pathomechanics, 417
 treatment, 417-19
 glenohumeral joint, posterior dislocation, 408
 glucosamine, 272-3
 glyceryl trinitrate (GTN) patches, 272
 golf (biomechanical aspects of injuries), 128-30
 hip pain, 130
 low back pain, 130
 shoulder pain, 129
 wrist pain, 128-9
 graded motor imagery and pain, 75
 granulomatous diseases, 83
 greater trochanteric pain, 622-3
 green tea, 273
 groin pain, 629-57
 acute groin injuries, 633, 642-3
 adductor-related, 643-50
 anatomy, 629-31
 assessment of severity, 635-9
 causes in athletes, 634
 classification, 633-4
 clinical examination, 635-9
 clinical overview, 634-42
 complete adductor avulsions, 654
 epidemiology, 631-2
 examination, 635-9
 hip adductors, 630
 hip flexors, 630
 iliopsoas-related, 650-1
 imaging, 639-42
 inguinal region, 631
 inguinal-related, 652-4
 injury prevention, 657
 longstanding groin pain, 633-4, 643-54
 nerve entrapments, 655

obturator neuropathy, 654-5
 PROM, 639
 pubic symphysis, 629-30
 pubic-related, 654
 referred pain to the groin, 656-7
 risk factors, 632
 stress fractures of the inferior pubic ramus, 656
 stress fractures of the neck of the femur, 655-6
 terminology and definitions, 632-4
 gymnast's wrist, 476

H

hallux limitus, 967
 hallux valgus, 967-8
 hamstring
 muscles, strengthening, 695-8
 proximal hamstring tendinopathy, 577-81
 proximal hamstring tendon rupture, 588-9
 tendon ruptures, 767
 upper hamstring tendinopathy, 710-11
 see also acute hamstring muscle strains
 hamstring injuries
 acute management phase, 692-3
 intramuscular tendon injuries, 690
 management, 690-703
 nerve entrapments, 711
 rehabilitation, 693
 subacute/conditioning phase, 694-701
 hamstring strains, prevention, 173-4, 705-7
 balance exercises/proprioception training, 705-6
 eccentric strength training, 705
 high-risk athlete, 706-7
 injury mechanisms, 173
 programs, 173-4
 risk factors, 173
 sport-specific training, 706
 hamstring synergists, strengthening for, 698-9
 hand injuries, 489-505
 Boutonnière deformity, 503-4
 carpometacarpal joint dislocations, 498-9
 chronic swan neck deformity, 503
 clinical approach, 489-92
 dislocation, 499
 exercises, 494
 finger joint dislocations, 499
 investigations, 492
 lacerations and infections, 504
 ligament and tendon injuries, 499-504
 mallet finger, 501-3
 metacarpal fractures, 494-7
 oedema control, 493-4
 overuse conditions, 504-5
 patient-reported outcome measures (PROMs), 493
 phalanges fractures, 497-8
 physical examination, 489-92
 sport-specific conditions, 490
 surgical referrals, 505
 taping and splinting, 494
 treatment principles, 492-4
 hand pain, 485-6
 HARM (heat, alcohol, running, massage), 250
 headache, 317-29
 cervicogenic, 322-8, 374
 classification, 317
 clinical approach, 318-20
 clinical measurement, 319-20
 external compression, 328
 high-altitude, 328
 hypercapnia, 329
 measurement, 319-20
 migraine, 320-2
 post-traumatic, 328
 primary, 320-2
 primary exercise, 322
 secondary, 322-9
 in sport, 317-18
 heel raise, single-leg, 102
 herniation of nucleus pulposus from intervertebral disc, 18
 high-altitude headache, 328
 high-intensity short-duration exercise, 190
 hip adductors, 630
 hip flexors, 630
 Hip and Groin Outcome Score (HAGOS), 232, 233
 hip
 control in functional task performance, 618-19
 instability, 614
 irritable, 984
 joint, 85
 lateral pain, 622-8
 ligaments, 596
 hip-related pain, 593-628
 acetabular labrum, 594-5
 adverse loading, 620-1
 arthroscopy, 621-2
 balance and proprioception, 618
 biomechanics, 593-8
 causes, 599
 chondral surfaces, 596
 chondropathy, 613-14
 clinical approach, 598-604
 diagnostic imaging tests, 605
 distal factors, 607

femoroacetabular impingement (FAI), 607-10
 functional anatomy, 593-8
 functional task performance, 620
 gait biodynamics, 620
 greater trochanteric pain, 622-3
 iliac crest pain, 623
 investigations, 603-4
 labral tears, 610-12
 ligamentum teres tears, 612
 morphology, 594
 muscle function, 596-8
 muscle strength, 616-18
 osteoarthritis, 610
 physical examination, 601-3
 predisposing factors, 604-7
 PROMs, 603-4
 proximal factors, 606
 rehabilitation, 614-15, 621-2
 remote factors, 606-7
 restoring ROM, 615
 surgical management, 621-2
 synovitis, 612-13
 systemic factors, 607
 treatment, 614-22, 625-8
 trunk muscle strength, 619-20
 homunculi, motor and sensory, 66
 hook of hamate fracture, 481-2
 hyaline cartilage, 14, 17-18
 hyaluronic acid, 269-70
 hypercapnia headache, 329
 hyperextension injuries (elbow), 460
 hypertrophy training, 149
 hyphema, 339

I

ice (acute injury management), 248, 692
 idiopathic scoliosis, 981
 iliac crest pain, 623
 iliopsoas-related groin pain, 650-1
 iliotibial band friction syndrome (IBTFS), 809-13
 imaging
 bone stress injuries, 35-7
 groin pain, 639-42
 neck pain, 353
 impingement syndromes, 481, 911
 inciting event (injury causation), 166
 individual needs
 sport and exercise medicine (SEM), 5
 individualisation (training), 140
 infections, 83
 inferential stimulation, 262
 inguinal hernia, 654
 inguinal region, 631

inguinal-related groin pain, 652-4
 diagnostic criteria, 652
 treatment, 652-4
 injection therapy, 268
 injections
 autologous blood, 273
 corticosteroid injection (lateral elbow tendinopathy), 448-9
 high-volume, 269
 local anaesthetic, 268-9
 mechanical, 269
 injured athlete
 from injury to return to play, 7
 injuries *see* sports injuries
 injury management, acute, 247-50
 compression, 248-50
 elevation, 250
 ice, 248
 initial management, 13-15
 optimal loading, 247-8
 pathophysiology, 13-15
 protection, 247
 injury prevention, 165-88
 ankle sprains, 174-7
 conceptual approach, 165-6
 hamstring strains, 173-4
 knee, 177-81
 and managing load, 183-6
 overuse injuries, 181-3
 playing surfaces, 187-8
 protective equipment, 187
 risk management, 166-73
 targeted program, 172-3
 input-dominated pain, 65-9
 from musculoskeletal tissues, 66
 neuropathic, 68-9
 referred pain, 66-8
 instability and injury, 212
 insurance medical assessment, 1014
 integrated performance health
 management and coaching model, 4
 intersection syndrome, 477-8
 interval training, 142
 intramuscular tendon injuries, 690
 iontophoresis, 448
 irritable hip, 984
 ischial bursitis, 711
 ischial tuberosity avulsion fracture, 589
 ischiofemoral impingement, 587
 isokinetic exercises
 for resistance, 146
 isometric exercises
 for resistance, 144-5
 tendinopathy rehabilitation, 50-1
 isotonic exercises
 for resistance, 145-6
 tendinopathy rehabilitation, 51

J

Jack's test for first metatarsophalangeal joint (MTPJ) range 97, 98, 101, 939
 'jersey finger', 504
 jet lag, 1023-5
 and air travel, 1023
 prevention, 1023-5
 symptomatic treatment for, 1025
 joint motion, lower limb, 85-8
 joints
 dislocation/subluxation, 14, 18-19
 overuse sports injuries, 43
 Jones fracture, 962-4
 Joplin's neuritis, 972

K

Kienböck's disease, 479-81
 kinetic chain integration, 433-5
 knee
 assessment, 102-3
 tibiofemoral alignment at the, 99
 see also anterior knee pain
 knee injuries, acute
 injury mechanisms, 177
 prevention, 177-81
 prevention programs, 177-81
 risk factors, 177
 knee joint, 85
 knee pain
 and cycling, 121-4
 and rowing, 133
 and swimming, 136
 see anterior knee pain; lateral knee pain; medial knee pain; posterior knee pain
 Köhler's disease, 956, 988-9

L

labral tears, 610-12
 lacerations and contusions (facial injuries), 332-3
 immediate management, 332
 management of larger, 332-3
 language and pain, 75
 laser therapy, 262-3
 lateral ankle pain, 929-33
 anterolateral impingement, 932-3
 causes, 929
 examination, 929
 peroneal tendinopathy, 929-31
 posterior impingement syndrome, 933
 referred pain, 933
 sinus tarsi syndrome, 931-2
 stress fractures of the talus, 933
 lateral collateral ligament complex (LCLC), 440

- lateral collateral ligament tears, 759-60
 - posterolateral corner (PLC) injuries, 759-60
 - lateral elbow pain, 136-7, 440-51
 - causes, 441, 451
 - examination, 441-2
 - history, 440-1
 - investigations, 442-3
 - lateral elbow tendinopathy, 443-51
 - acupuncture, 447
 - autologous blood, 450
 - botulinum toxin, 450
 - bracing and taping, 447-8
 - clinical features, 444-5
 - clinical relevant pathology, 443-4
 - correct predisposing factors, 450
 - corticosteroid injection, 448-9
 - electrotherapeutic modalities, 446
 - exercises for strengthening and coordination, 445-6
 - iontophoresis, 448
 - nitric oxide donor therapy, 449-50
 - return to activity, 450-1
 - soft tissue therapy, 446-7
 - surgery, 450
 - treatment, 445
 - trigger points, 447
 - lateral exertional compartment syndrome, 843-4
 - lateral femoral cutaneous nerve injury ('meralgia paresthetica'), 676
 - lateral hip pain, 622-8
 - lateral knee pain, 805-16
 - causes, 806, 814-16
 - clinical approach, 806-9
 - iliotibial band friction syndrome, 809-13
 - investigations, 809
 - lateral meniscus
 - abnormality, 813-14
 - lateral ligament injuries, 899-900
 - treatment and rehabilitation, 900-4
 - lateral meniscus abnormality, 813-14
 - lateral plantar nerve, 947-8
 - leg length assessment, 99-100
 - leg pain, 825-46
 - biomechanics, 826
 - causes, 827, 829
 - chronic exertional compartment syndrome, 834-6, 841-45
 - clinical approach, 825-36
 - diagnostic questions, 830
 - fractured tibia and fibula, 846
 - investigations, 832-6
 - medial tibial stress syndrome, 839-41
 - nerve entrapment, 845-6
 - outcome measures, 832
 - patient-reported outcome measures, 833
 - periosteal contusion, 846
 - physical examination, 828-32
 - referred pain, 845
 - stress fracture of the anterior cortex of the tibia, 838
 - stress fracture of the fibula, 845
 - stress fracture of the tibia, 836-8
 - vascular entrapments, 846
 - leisure activities and injury, 212
 - lens dislocation (eye injury), 339
 - ligament injuries
 - bifurcate, 952
 - classification of, 20
 - dorsal calcaneocuboid, 952
 - examination, 215
 - hand and finger, 499-504
 - hips, 596
 - medial (deltoid), 904-5
 - and NSAIDs, 266
 - overuse sports injuries, 43
 - sprain/tear, 14, 19-21
 - see also* lateral ligament injuries
 - ligamentum teres tears, 612
 - lisfranc joint injuries, 952-5
 - causes, 953
 - clinical features, 953-4
 - investigations, 954
 - treatment, 954-5
 - load management
 - for performance enhancement, 150-1
 - and prevention of injury, 183-6
 - load ratio, monitoring acute:chronic, 184-6
 - local anaesthetic injections, 268-9
 - local anaesthetic patches, 267
 - long-duration exercise, 190-1
 - long thoracic nerve injury, 430
 - low back pain
 - and cycling, 123
 - and rowing, 130-2
 - lower limb biomechanical abnormalities
 - biofeedback, 108-9
 - foot orthosis, 109-12
 - management of, 108-14
 - movement pattern retraining, 108-9
 - taping, 112-14
 - lower limb biomechanics, 85-90
 - common observations, 101
 - conditions related to suboptimal, 106-8
 - ideal neutral stance position, 88-90
 - joint motion, 85-8
 - lower limb biomechanics assessment, 95-105
 - dynamic movement, 104-5
 - functional tests, 100-4
 - structural ('static') assessment, 96-100
 - summary, 105
 - lower limb joint motion, 85-8
 - ranges of motion, 86
 - lower limb overuse injuries, 107-8
 - lower limb stiffness, 749
 - lower limb tests, functional, 100-4
 - landing-specific considerations, 103-4
 - single-leg heel raise, 102
 - single-leg squat, 102-3
 - single-leg stance with progressions, 100
 - low-level laser therapy (LLLT), 262-3
 - lubricants (soft tissue therapy), 255
 - lumbar spine, referred pain from, 576, 707-9
 - lumbopelvic control, 155-6
 - lumbopelvic dysfunction, 154-5
 - lumbopelvic pain, 162
 - lunotriquetral dissociation, 482-3
- M**
- magnetic resonance imaging (MRI), 12, 35-7, 129, 221-3, 224-7, 306, 353, 469
 - acute knee injuries, 722-3
 - groin pain, 641-2
 - shoulder investigations, 401
 - maisonneuve fracture, 905
 - mallet finger, 502
 - management
 - bone stress injuries, 38-42
 - of concussion, 298-310
 - of mechanical neck pain, 365-74
 - and return-to-play decisions, 292
 - sport and exercise medicine (SEM), 4-5
 - sports injuries, 13-15
 - mandibular fractures, 344-5
 - manual treatments (musculoskeletal conditions), 250-9
 - joint techniques, 251-2
 - neurodynamic techniques, 258-9
 - soft tissue therapy, 252-8
 - manipulation (musculoskeletal conditions), 251-2
 - Marfan syndrome, 79, 82, 340
 - massage and recovery, 191-2, 200
 - maxillary fractures, 344
 - Maximal Aerobic Speed (MAS) training, 143
 - mechanotherapy, 241-5
 - medial ankle pain, 917-29
 - causes, 917, 927
 - examination, 918
 - flexor hallucis longus tendinopathy, 924-6
 - history, 917-18
 - investigations, 918-19

- medial calcaneal nerve entrapment, 928-9
- medial malleolar stress fracture, 928
- PROMs, 918, 920-23
- tarsal tunnel syndrome, 926-8
- tibialis posterior tendinopathy, 923-4, 955
- medial calcaneal nerve entrapment, 928-9
- medial collateral ligament (MCL)
 - injury, 729-36, 816-17
 - chronic, 819-20
 - rehabilitation, 730-1
 - treatment, 729-36
- medial (deltoid) ligament injuries, 904-5
- medial knee pain, 136, 816-20
 - causes, 817, 819-20
 - chronic, 819-20
 - medial meniscus abnormality, 817-18
 - patellofemoral syndrome, 817
 - Pellegrini-Stieda syndrome, 819-20
 - pes anserinus tendinopathy/bursitis, 819
- medial malleolar stress fracture, 928
- medial meniscus abnormality, 817-18
- medial tibial stress syndrome, 839-1
- medial tubercle fractures, 911
- median nerve entrapment, 461
- medical assessment, periodic (sports injuries), 171-2
- medical equipment, 1018
- medicine team
 - attributes, 1017-18
 - equipment, 1018
 - indemnity and trauma training, 1018
 - support team, 1017
 - sport and exercise, 2-3
- meniscal injuries, 723-8
 - clinical features, 724-6
 - non-surgical management, 727
 - rehabilitation, 727-8
 - surgical management, 727
 - symptomatic meniscus tears, 748
 - treatment, 726-7
- meniscal lesions, degenerative, 724-5
- mental health issues and concussion, 310
- metacarpal fractures, 494-7
- metatarsal, avulsion fracture of the base of the fifth, 908-10
- midfoot pain, 948-58
 - abductor hallucis strains, 956
 - causes, 948, 955-8
 - cuboid syndrome, 956
 - extensor tendinopathy, 956
 - first tarsometatarsal joint pain, 955-6
 - investigations, 949
 - Köhler's disease, 956
 - lisfranc joint injuries, 952-5
 - midtarsal joint sprains, 952
 - navicular stress fracture, 949-52
 - peroneal tendinopathy, 956
 - stress fracture of the cuboid, 956
 - stress fracture of the cuneiforms, 956
 - tarsal coalition, 957-8
 - tibialis posterior tendinopathy (distal pain presentation), 955
- midportion Achilles tendinopathy, 872-84
 - biomechanical factors, 877
 - cell-based therapies, 883
 - four-stage program, 878
 - pathology, 872-6
 - predisposition factors, 876-7
 - sclerosing, 882-3
 - treatment, 877-84
- midtarsal joint sprains, 952
- migraine, 320-2
 - post-traumatic, 328
- military personnel, 991-1001
 - aerobic fitness, 997-8
 - blister injuries, 994-5
 - body composition, 998
 - common injuries, 991-6
 - epidemiology of military injuries, 992-3
 - gender, 997-8
 - injury prevention, 996-1001
 - injury surveillance, 996-7
 - overuse of lower limb, 994
 - parachute injuries, 995
 - previous injuries, 998-9
 - special culture, 991-2
 - warm-up/stretching, 1000-1001
 - weekly running distance, 999-1000
- mobilisation (musculoskeletal conditions), 251-2
- Morton's interdigital neuroma, 970-1
- motion
 - lower limb joint, 85-8
- motor adaption to pain, 72-5
 - education, 74
 - exercise/progressive loading, 74-5
 - graded motor imagery, 75
 - manual therapy, 74
 - medications, 74
 - treatment options for patients, 72-3
- motor control
 - balance between movement and stiffness, 155
 - and biology of pain, 158-9
 - clinical principles, 160-4
 - common misconceptions, 155
 - and core stability, 154-64
 - and the individual, 159
 - lumbopelvic control, 155-6
 - lumbopelvic dysfunction, 154-5
 - lumbopelvic pain, 162
 - and motor learning, 157-8
 - and neck pain, 370-2
 - and treatments for musculoskeletal conditions, 245-6
 - and pain prevention, 164
 - and rehabilitation, 156-7
 - motor function (neck pain), 370-73
 - motor learning, 163
 - mouth guards, 341-3
 - movement assessment, dynamic, 104-5
 - movement biomechanics (running), 90-5
 - movement pattern retraining (lower limb biomechanics), 108-9
 - MRI *see* magnetic resonance imaging (MRI)
 - MTP joint synovitis, 965-6
 - Müller-Wohlfahrt, Hans-Wilhelm, 1033
 - multiskilling, 3
 - muscle cramps, 25, 849
 - muscle disorders, 81
 - muscular endurance, 149
 - muscle factors
 - bone stress injuries, 33
 - muscle function in hips, 596-8
 - muscle function tests (neck pain), 358-62
 - muscle injuries
 - acute, classification, 22
 - acute compartment syndrome, 24-5
 - chronic compartment syndrome (CCS), 45
 - contusion, 23-4
 - cramp, 25
 - exercise-induced muscle soreness, 45-6
 - gastrocnemius muscle strains, 853-6
 - and mechanotherapy, 245
 - myofascial pain, 43-5
 - myositis ossificans, 24
 - and NSAID, 266
 - overuse sports injuries, 43-6
 - shoulder, 429
 - soleus muscle strains, 856-9
 - strain/tear 14, 21-3
 - muscle soreness and NMES, 193
 - muscle weakness following ACL reconstruction, 749-50
 - musculoskeletal conditions, management of, 976-91
 - acute fractures, 976-9
 - overuse injuries of the physis, 979
 - musculoskeletal conditions, treatments for, 239-75
 - actovegin, 269
 - acute injury management, 247-50
 - analgesics, 264

- autologous blood, 273
 - bisphosphonates, 272
 - cell therapy, 274
 - chondroitin, 272-2
 - corticosteroid, 270-1
 - dry needling, 256-8
 - electromagnetic therapy, 263
 - electrophysical agents, 261-4
 - exercise-induced hypoalgesia, 246
 - extracorporeal shockwave therapy, 263-4
 - glucosamine, 272-3
 - green tea/polyphenols, 273
 - GTN patches, 272
 - hyaluronic acid, 269-70
 - interferential stimulation, 262
 - laser, 262-3
 - local anaesthetic
 - injections, 268-9
 - manual treatments, 250-9
 - mechanical and high-volume
 - injections, 269
 - mechanotherapy, 241-5
 - motor-control training, 245-6
 - neurodynamic techniques, 258-9
 - neuromuscular stimulators, 262
 - neuropathic pain, 267-8
 - neutraceuticals in injury
 - management, 272-3
 - NSAIDs, 264-7
 - omega-3 fatty acids, 272-3
 - platelet-rich plasma, 273
 - prolotherapy, 269
 - quinolone antibiotics, 272
 - sclerosant, 269
 - sleep medication, 271
 - soft-tissue therapy, 252-6
 - surgery, 274-5
 - taping, 259-61
 - TENS, 262
 - therapeutic exercise, 241-6
 - therapeutic medication, 264-72
 - therapeutic ultrasound, 261
 - traumeel, 269
 - vitamin D, 273
 - musculoskeletal symptoms and
 - diagnosis, 212
 - musculoskeletal tissues
 - and input-dominated pain, 66
 - myofascial, 574-6
 - myofascial pain syndrome, 43-5, 68
 - myofascial tension, sustained, 254
 - myositis ossificans, 24, 670-1
- N**
- nasal fractures, 334
 - natural grass *vs.* artificial turf, 187-8
 - navicular stress fracture, 949-52
 - clinical features, 949-50
 - investigations, 950
 - management, 950-52
 - neck extensors, 372
 - neck flexors, 370-1
 - neck pain, 347-75
 - acceleration/deceleration injury (whiplash), 375
 - active movement tests, 355-7
 - acute wry neck, 374
 - anatomical considerations, 347-9
 - axio-scapular muscles, 361-2, 372
 - biopsychosocial model for, 350
 - brachial plexus injuries (stingers/burners), 375
 - cervical arterial dysfunction, 364
 - cervical extensors, assessment of, 360-1
 - cervical flexors, assessment of, 358-60
 - cervical nerve injury, 374-5
 - cervical radiculopathy, 374-5
 - cervical sensorimotor function, tests for, 362-4
 - cervicogenic headache, 374
 - clinical neurological examination, 357
 - clinical perspective, 349-65
 - conditions, 374-5
 - education, 366-7
 - imaging, 353
 - maintenance program, 374
 - manual therapy, 367-9
 - mechanical, management of, 365-74
 - motor control, 370-2
 - muscle function, tests of, 358-62
 - nervous system, tests of the, 357-8
 - neural tissue mobilisation, 369-70
 - pain management, 367
 - passive tests, 357
 - patient reported outcome measures (PROMs), 352
 - performance-based outcome measures, 364-5
 - physical examination, 353-64
 - postural analysis, 354-5
 - posture, 371-2
 - presentations (red flags), 351
 - psychological features, 352
 - quantitative sensory testing, 357
 - resistance training, 372-3
 - social/sport factors, 352
 - sport and functional modifications, 365-6
 - symptoms and behaviours, 351
 - training motor function, 370-73
 - training sensorimotor control, 373-4
 - nerve entrapments
 - calf pain, 862-3
 - compartment syndrome, 844
 - hamstring, 711
 - groin pain, 655
 - medial calcaneal, 928-9
 - nerve injuries, 26, 52
 - cervical, 374-5
 - nerve tissue pain (neuropathetic pain), 68-9, 268
 - nervous system tests (neck pain), 357-8
 - neural mechanosensitivity, 214
 - neural tissue mobilisation, 369-70
 - neurodynamic techniques (musculoskeletal conditions), 258-9
 - neuromechanical sensitivity and
 - injuries, 215-19
 - neuromuscular control exercises, 699-701
 - neuromuscular electrical stimulation (NMES), 193, 199, 262
 - neuromyofascial causes of calf pain, 862
 - neuropathic pain, 68-9
 - neuropraxia, 26
 - neuropsychological tests (cognitive function and concussion), 308-9
 - neurovascular injuries, 429-32
 - neutral stance position, 88-90
 - nociception, 56-9, 67
 - non-steroidal anti-inflammatory drugs (NSAIDs), 264-7
 - adverse effects, 266-7
 - hamstring injuries, 692-3
 - use in sport, 265-6
 - and musculoskeletal injury, 266
 - and wrist injuries, 486
 - Nordic hamstring, 699
 - nose injuries, 333-5
 - fractures, 334
 - septal haematoma, 334-5
 - nosebleed, 333-4
 - nutraceuticals in injury management, 272-3
 - nutrition and recovery, 196-9, 200
 - nutritional factors and injury, 213
 - nutritional strategies
 - and prevention of stress fractures, 182
- O**
- obturator neuropathy, 654-5
 - olecranon fractures, 456-7
 - omega-3 fatty acids, 272-3
 - on-field management of concussion, 298-300
 - on-field rehabilitation, 8
 - onychocryptosis 972
 - open chain exercises, 146-8

open wounds, treatment of, 27
 optimal loading (acute injury management), 247-8
 orbital fracture, 343-4
 orthoses, 109-12, 183, 788
 Osgood-Schlatter disease, 802, 888, 984-6
 osteoarthritis, 610, 762-3, 814
 osteochondral injuries, 17-18
 osteochondral lesions of the talar dome, 906-8
 otitis externa, 336
 Ottawa ankle rules, 897, 898, 905
 overload (training), 140
 overtraining and injury, 213
 overuse injuries *see* sports injuries, overuse

P

pain, 55-62

and the brain, 59-61
 biology of, 158-9
 centrally dominated, 69-75
 definition, 55-6
 and diagnosis, 211-12
 and language, 75
 persistent, 60-1, 154
 prevention, 164
 and PROMs, 231-7
 radicular, 67-8
 referred, 66-8, 576, 656-7, 677, 707-9, 816, 845, 888, 933
 syndromes, 83-4
 treating, 61-3
 treatment options, 72-5
 from trigger points, 44, 83
see also individual locations and types of pain

pain, clinical aspects, 65-76
 centrally dominated, 69-72
 input-dominated, 65-9

palpation and injuries, 215

paratenonitis, 49-50

passive movement tests, 357

patellar

acute patellar trauma, 762-5
 dislocation, 762-5
 fracture, 762
 pain, 748-9
 tendinopathy, 769-1, 791-9, 986
 tendon rupture, 765-7
 treatment, 765

patellofemoral instability, 792-3

patellofemoral pain, 769, 770, 771, 774-92

patellofemoral pain, treatment, 782-90
 braces, 789
 exercise, 783-90
 extrinsic contributing factors, 790

in-shoe foot orthoses, 788
 manual therapies, 788
 physical intervention, 789-90
 surgery, 790

patellofemoral syndrome, 817

pathophysiology, 30-31
 and concussion, 298

patient-reported outcome measures (PROMs), 210, 231-8

Achilles pain, 872
 anterior knee pain, 774-5
 considerations for 'good' use, 232-5
 definition, 231
 easiness to use, 232-3
 evaluation dimensions, 233
 groin pain, 639
 hand and finger injuries, 493
 hip-related pain, 603-4
 leg pain, 833
 low back pain, 529-30
 medial ankle pain, 918, 920-3
 and neck pain, 352
 patellofemoral pain, 775
 and patient changes, 233-5
 plantar fasciopathy, 941-6
 use in sports medicine, 231-2
 wrist pain, 471

pectoralis major tears, 429

Pellegrini-Stieda syndrome, 819-20

pelvis/hip function assessment, 102

perforated eardrum, 335-6

performance enhancement, training
 load management for, 150-1

periodic medical assessment (PMA), 171-2

periodisation (training), 139-40

periosteal contusion, 16-17

peroneal tendinopathy, 929-33, 956

persistent pain, 60-1
 contribution of body and mind, 154

Perthes' disease, 983

pes anserinus tendinopathy/bursitis, 819-20

phalanges fractures, 497-8

pharmacotherapy, 799

physical activity and clinician, 6

physical activity history

bone stress injuries, 34

physical conditioning

bone stress injury management, 40

physical examination

Achilles pain, 868-72

acute Achilles tendon rupture, 890

acute knee injuries, 716-17

anterior knee pain, 771-4

anterior thigh pain, 661-4

bone stress injuries, 35

gastrocnemius muscle strains, 853

hip-related pain, 601-3

and injury, 213-21

and leg pain, 828-32
 posterior thigh pain, 682-4
 shoulder pain, 383-8
 and wrist pain, 466

physical examination and neck pain, 353-65

active movement tests, 355-7

anterior knee pain, 771-4

axio-scapular muscles, 361-2

cervical arterial dysfunction, 364

cervical extensors assessment, 360-1

cervical flexors assessment, 358-60

cervical sensorimotor function, 362-4

clinical neurological examination, 357

manual examination of segmental dysfunction, 357

muscles function, 358-62

nervous system, 357-8

passive tests, 357

patellofemoral pain, 775

postural analysis, 354-5

quantitative sensory testing, 357

Physiotherapy Evidence Database (PEDro), 10, 11-12

piriformis syndrome, 585-7

plantar fasciopathy, 941-6

alignment of the foot, 942

clinical features, 942-3

flexibility and strength, 942

investigations, 942-3

PROMs, 940

risk factors, 941-3

standing work, 942

training volume, 942

treatment, 943-6

weight and BMI, 941

plantar nerve, first branch of lateral (Baxter's nerve) entrapment, 947-8

plantar plate tear, 972-3

plantar warts, 971

platelet-rich plasma (PRP), 273

play, return to (RTP), 285-93

StARRT, 285-93

playing surface

bone stress injuries, 33

POLICE (protection, optimal loading,

ice, compression, elevation),

247, 280

polyphenols, 273

popliteal artery entrapment syndrome (PAES), 859-61

popliteal cysts, 986

popliteus tendinopathy, 823

posterior cruciate ligament (PCL) tears, 755-9

clinical features, 756-7

treatment, 757-9

- posterior dislocation (acute elbow injuries), 457-9
 - posterior knee pain, 820-24
 - Baker's cyst, 822-3
 - biceps femoris tendinopathy, 823
 - causes, 821, 824
 - claudication, 824
 - clinical evaluation, 820-22
 - deep venous thrombosis, 824
 - examination, 821-2
 - gastrocnemius tendinopathy, 824
 - investigations, 822
 - popliteus tendinopathy, 823
 - posterior impingement syndrome, 887-8, 933
 - posterior thigh compartment syndrome, 587-8
 - posterior thigh pain, 679-712
 - acute hamstring muscle strain, 685-705
 - adductor magnus strains, 711
 - avulsion of the hamstring from the ischial tuberosity, 709-10
 - causes, 682
 - clinical approach, 681-5
 - compartment syndrome, 711
 - functional anatomy, 679-81
 - hamstring strains prevention, 705-7
 - investigations, 684-5
 - ischial bursitis, 711
 - lumbar spine, 708-9
 - nerve entrapments, 711
 - physical examination, 682-5
 - referred pain, 707-9
 - sacroiliac complex, 709
 - trigger points, 707
 - upper hamstring tendinopathy, 710-11
 - vascular, 712
 - posterolateral corner (PLC) injuries, 759-60
 - post-traumatic headache, 328
 - post-traumatic migraine, 328
 - post-traumatic synovitis, 913-14
 - postural abnormalities and young athlete, 982
 - postural imbalance, 514
 - posture, 371-2
 - Pott's fracture, 905
 - power training, 148-9
 - pregabalin, 267
 - Pre-Hospital Emergency Care Course (PHECC), 299
 - Pre-Hospital Trauma Life Support (PHTLS), 299
 - PRICE (protection, rest, ice, compression and elevation), 247, 280
 - primary headache, 320-2
 - primary exercise headache, 322
 - Profile of Mood State (POMS) scores, 191
 - prolotherapy, 269
 - Proprioceptive Neuromuscular Facilitation (PNF) stretching, 150
 - protective equipment and injury prevention, 187
 - protection (acute injury management), 247
 - proximal hamstring tendinopathy, 577-81
 - proximal hamstring tendon rupture, 588-9
 - psychological effects
 - of active recovery, 191
 - of massage, 192
 - psychological factors and injury, 213
 - pubic ramus stress fractures, 590
 - pubic related groin pain, 654
 - pubic symphysis, 629-30
 - published appraisals and (EBP), 10-11
 - PubMed, 10, 11
 - pulsed electromagnetic field therapy (PEMF), 263
- Q**
- quadriceps contusion, 664-71
 - compartment syndrome, 670
 - complications, 670-1
 - distinguishing features, 665
 - grading, 665
 - myositis ossificans, 670-1
 - treatment, 666-770
 - quadriceps strain, 671-4
 - avulsion injury, 673-4
 - complete rectus femoris tear, 673
 - distal quadriceps muscle strain, 671-3
 - prevention, 674
 - proximal rectus femoris strain, 672-3
 - treatment, 672
 - quadriceps tendinopathy, 802
 - Quality Assessment of Diagnostic Accuracy Studies (QUADAS), 206
 - quantitative sensory testing (neck pain), 357
 - quinolone antibiotics, 272
- R**
- radial column presentations, 470-8
 - radial epiphyseal injury (gymnast's wrist), 476
 - radial head fracture, 456
 - radial sensory nerve compression, 478
 - radial tunnel syndrome, 460-1
 - radicular pain, 67-8
 - radiological investigation, 466-70
 - range of motion exercises and neck pain management, 368-9
 - range of motion testing
 - active, 214
 - passive, 215
 - randomised control trials (RCTs), 10, 11
 - rearfoot pain, 937-51
 - calcaneal stress fractures, 947
 - causes, 939
 - clinical features, 942-3
 - examination, 939
 - fat pad contusion, 946-7
 - first branch of lateral plantar nerve (Baxter's nerve) entrapment, 948-9
 - history, 937
 - inferior foot pain, 939
 - investigations, 939-40
 - plantar fasciopathy, 941-6
 - PROMs, 940
 - treatment, 943-6
 - recovery, 189-200
 - active, 189-91, 200
 - assessment, 189
 - compression, 195, 200
 - massage, 191-2, 200
 - modalities, 200
 - neuromuscular electrical stimulation (NMES), 193, 200
 - nutrition, 196-9, 200
 - practical considerations, 199-200
 - sleep, 194-5
 - stretching, 193-4, 200
 - water immersion, 195-6, 200
 - rehabilitation, 239
 - Achilles tendon ruptures, 890-2
 - ACL injury, 745-50
 - acute ankle injuries, 900-4, 908
 - closed chain exercises, 147-8
 - effective planning, 278
 - and the flexibility process, 150
 - goal setting, 279-80
 - groin pain, 649
 - hamstring muscle strains, 693
 - hip-related pain, 614-15, 621-2
 - isometric exercises, 50-1
 - isotonic exercises, 51
 - lateral ligament injuries, 900-4
 - low back pain, 563, 748
 - lower limb (tendinopathy) 50-1
 - medial collateral ligament injury, 730-1
 - meniscal injuries, 727-8
 - open chain exercises, 147-8
 - phases, 280-3
 - principles of sports injury, 277-83
 - and return to play (RTP), 285-93
 - scapular dyskinesis, 420-4
 - shoulder injuries, 402-7, 412-14
 - and sleep, 271

- sports, 1029
 - targeted interventions, 279-80
 - tendon injuries, 145
 - when it doesn't go to plan, 283
 - wrist, 475, 487
 - rehabilitation gym, 8
 - research and (EBP), 10-12
 - resistance exercises
 - classification, 146-8
 - closed chain exercises, 146-8
 - isokinetic exercises, 146
 - isometric exercises 144-5
 - isotonic exercises, 145-6
 - open chain exercises, 146-8
 - types of, 144-6
 - resistance training, 144-9
 - hypertrophy training, 149
 - loading patterns, 148
 - muscular endurance, 149
 - and neck pain, 372-3
 - power, 148-9
 - qualities of, 148-9
 - strength, 148
 - rest (acute injury management), 692
 - retinal detachment, 340
 - retinal haemorrhage, 339
 - retrieving articles and EDP, 11
 - retrocalcaneal bursitis
 - (enthesus organ), 884-7
 - return to play (RTP), 285-93
 - ACL injury, 751-2
 - acute Achilles tendon rupture, 889-2
 - acute ankle injuries, 902-3
 - additional perspectives, 289
 - beyond risk for injury, 287
 - and concussion, 306-10
 - decision making, 290-3
 - gastrocnemius muscle strains, 856
 - lateral elbow tendinopathy, 451
 - outcomes and probabilities, 288-9
 - and PMA, 1014
 - risk tolerance modifiers, 287
 - and shoulder injuries, 437-8
 - StARRT, 285-93
 - tissue health, 286-7
 - tissue stress, 287
 - return-to-sport phase
 - groin pain, 649
 - posterior thigh pain, 701-3
 - reverse Nordics, 669
 - rheumatological conditions, 80-1
 - RICE (rest, ice, compression, elevation) 692
 - rib stress fractures (RFS), 132, 515
 - rib trauma, 516-7
 - risk factors
 - acute hamstring pain, 703-5
 - bone stress injuries, 31-4
 - groin injuries, 632
 - plantar fasciopathy, 941-3
 - risk management and injury
 - prevention, 166-73
 - Roland Morris Disability Questionnaire, 529-30
 - rotator cuff
 - injuries, 402-7
 - tears, 406-7
 - tests, 390
 - rotator cuff tendinopathy, 402-6
 - rowing (biomechanical aspects of injuries), 130-3
 - chest wall pain, 132
 - knee pain, 133
 - low back pain, 130-2
 - wrist and forearm injuries, 132-3
 - running
 - barefoot/minimalist, 183
 - running (bone stress injury management), 40-2
 - anti-gravity treadmill training, 41
 - gait retraining, 41-2
 - program design, 42
 - returning to, 40-2
 - running (movement biomechanics), 90-5
 - angle and base of gait, 93
 - comparing heel and forefoot strike patterns, 94-5
 - influence of fatigue, 95
 - influence of gait velocity, 94
 - initial swing, 92-3
 - landing point relative to centre of mass, 94
 - loading (heel strike to foot flat), 90-1
 - midstance (foot flat to heel off), 91-2
 - propulsion (heel off to toe off), 92
 - terminal swing, 93
- S**
- scaphoid fracture, 472-5
 - scapholunate dissociation, 479
 - scapular
 - involvement tests, 390
 - in normal shoulder function, 379-80
 - snapping scapular syndrome, 432-3
 - scapular biomechanics
 - abnormal, 118
 - normal biomechanics of the scapular, 117-18
 - in shoulder injuries, 118-19
 - scapular dyskinesis, 419-24
 - rehabilitation, 421-4
 - and shoulder pain, 419-20
 - sacral and pubic ramus stress fractures, 590
 - sacroiliac complex, 709
 - sacroiliac joint dysfunction, 581-5
 - SCAT3™, 301-4, 312-15
 - Scheuermann's disease, 512
 - sclerosant, 269
 - sclerosing, 882-883
 - second impact syndrome (SIS), 310
 - secondary headache, 322-9
 - self-treatment of soft tissues, 256
 - sensation of primary nociceptors ('peripheral sensitisation'), 57-8
 - sensation of spinal nociceptors ('central sensitisation'), 58-9
 - sensorimotor control (neck pain), 373-4
 - sensory discrimination training and pain, 75
 - septal haematoma, 334
 - serotonin reuptake inhibitors, 267
 - sesamoid injuries, 968-9
 - Sever's disease, 888, 988
 - sex and periodic medical assessment, 1012
 - sexual activity and teams, 1020
 - shin pain, common causes, 827
 - 'shin splints' 4
 - shoes and inserts
 - bone stress injuries, 33
 - patellofemoral pain, 788
 - and preventing overuse injuries, 182-3
 - shoulder injuries
 - anterior dislocation, 408
 - Bankart repair, 408
 - fractures, 425, 432
 - neurovascular injuries, 429-32
 - and overhead athletes, 433-8
 - posterior dislocation of the glenohumeral joint, 408
 - rehabilitation guidelines, 402, 412-14
 - and return to play, 437-8
 - rotator cuff injuries, 402-7
 - scapular biomechanics in, 118-19
 - and tennis, 137-8
 - treatment, 402-7
 - shoulder instability, 407-14
 - acquired sport-specific instability (AIOS), 408-12, 418
 - atraumatic multidirectional instability (AMBRI), 411
 - rehabilitation, 402, 412-14
 - traumatic shoulder instability (TUBS), 407-8
 - shoulder pain, 377-438
 - acromioclavicular joint injuries, 426-8
 - acute *vs.* overuse, 380
 - adhesive capsulitis 'frozen shoulder' 429

- anatomy, 377-80
- biceps-related pathology, 414-16
- biomechanics, 377-80
- causes, 380-2, 428-323
- clavicle fractures, 424-6
- dynamic stabilisers, 379
- and golf, 129-30
- impingement, 380-2, 388-9
- muscle tears, 429
- neurovascular injuries, 429-32
- objective measurements, 397-9
- patient-reported outcome measures, 400-1
- pathological glenohumeral internal rotation deficit, 417-19
- scapular dyskinesis, 419-24
- scapular in normal shoulder function, 379-80
- SLAP lesions, 394, 414-17
- special considerations for the overhead athlete, 433-8
- static stabilisers, 377-9
- and swimming, 133-6
- and tennis, 137-8
- young athletes, 979-80
- shoulder pain, clinical approach, 382-402
 - additional tests, 396
 - arthrography, 401
 - biceps pathology, 394
 - clinical evaluation of GIRD, 394-5
 - diagnostic testing, 388-96
 - diagnosis arthroscopy, 402
 - history, 383
 - impingement tests, 388-90
 - instability tests, 390
 - investigations, 397-402
 - MRI, 401
 - physical examination, 383-8
 - rotator cuff tests, 390
 - scapular involvement tests, 390
 - screening of the kinetic chain, 396
 - shoulder symptom modification procedure, 395-6
 - SLAP lesion tests, 394
 - ultrasound, 401
 - X-ray, 397-401
- shoulder symptom modification procedure (SSMP), 395
- side strain, 519-20
- side-lying slump test, 664
- Sinding-Larsen-Johansson disease, 986
- Sinding-Larsen-Johansson lesion, 802
- single-leg heel raise (functional lower limb test), 102
- single-leg squat (functional lower limb test), 102-3
- single-leg stance with progressions (functional lower limb test), 100
- sinus tarsi syndrome, 914, 931-2
- skill training
 - as agility, 144
 - as speed, 143-4
- skin cancers, 52-3
- skin infections, 52-3
- skin injuries, 14, 27
 - overuse sports injuries, 52-3
- SLAP lesions, 380, 382, 394, 402, 411, 414-17
 - treatment, 416-17
- sleep and recovery, 194-5, 200
- sleep hygiene strategies, 195
- sleep medication, 271
- slipped capital femoral epiphysis, 983-4
- snapping scapular syndrome, 432
- soccer and groin pain, 631
- social/sport factors (neck pain), 352
- soft tissue
 - injuries, 16
 - laxity, 749
 - stiffness, 749
 - tumours, 78-80
- soft tissue therapy, 252-6
 - combination treatments, 255
 - depth of treatment, 255
 - digital ischaemic pressure, 253-4
 - friction, 254-5
 - lubricants, 255
 - self-treatment, 256
 - sustained myofascial tension, 254
 - treatment position, 252-6
 - vacuum cupping, 255
- soleus muscle strains, 856-9
- somatic pain, 68
- specificity (training), 140
- speed intervals, 143
- speed training, 143-4
 - skill training as speed, 144
 - technique training, 143
- spinal cord and pain, 60
- spinal examination, 214, 219-21
- spiral fracture (distal third), 965
- spondyloarthropathies, 589-91
- spondylolisthesis, 543-5, 981-2
- spondylolysis, 981
- spondylolytic bone stress injury, 541-2
- Sport Concussion Assessment Tool (SCAT3)TM
 - Child-SCAT3TM, 312-15
 - SCAT3TM, 301-4, 311
- sport and exercise medicine (SEM), 2-8
 - case manager approach, 6-8
 - challenges of management, 4-5
 - clinician, 5-6
 - coach, the athlete and the clinician, 5-6
 - from injury to return to play, 7
- model, 3-4
- team, 2-4
- sports equipment
 - and preventing overuse injuries, 183
- sports injuries
 - circumstances, 211
 - classification, 14
 - conditions that masquerade as, 77-84
 - cryotherapy modalities used to treat, 249
 - ear, 335-6
 - and equipment, 213, 221
 - examination, 213-21
 - eye, 336-40
 - facial, 333, 343-5
 - and gender, 166
 - HARM, 250
 - headache, 317-18
 - history, 211-13
 - ligament testing, 215
 - medical assessment, periodic, 171-2
 - medical therapies, 692-3
 - muscle activation, 692
 - nose, 333-5
 - and pain, 211-12
 - palpation, 215
 - POLICE, 247
 - prevention models, 166-73
 - prevention of re-injury, 282-3
 - PRICE, 247
 - range of motion testing, 214-15
 - RICE, 692
 - season analysis, 170-1
 - soft tissue, 16
 - spinal examination, 219-21
 - and strength testing, 215
 - and surgery, 274-5
 - surveillance program, 168
 - swelling, 212
 - and technique, 221
 - teeth, 340-3
 - see also* injury management, acute; injury prevention; rehabilitation
- sports injuries, biomechanical aspects
 - in specific sports, 121-38, 166
 - cricket fast bowling, 125-7
 - cycling, 121-4
 - golf, 128-30
 - rowing, 130-3
 - swimming, 133-6
 - tennis, 136-8
- sports injuries, overuse, 29-53
 - barefoot/minimalist running, 183
 - biomechanical abnormalities, 214
 - bone stress injury, 29-43
 - bursa, 51-2
 - hand and finger, 505
 - joint, 43
 - ligament, 43

- lower limb, 107-8, 183
 - muscle, 43-6
 - nerve, 52
 - nutritional strategies to prevent
 - stress fractures, 182
 - orthoses, 183
 - overuse injuries of the physis, 979
 - predisposing factors, 29
 - prevention, 181-3
 - shoe selection, 182-3
 - skin, 52-3
 - sports equipment, 183
 - stretching, 181-2
 - structured training programs, 182
 - technique modification, 182
 - tendon (tendinopathy), 46-51
 - sports medicine and PROMs,
 - 210, 231-8
 - sports surfaces and injury prevention,
 - 187-8
 - sport-specific assessment, 105
 - sport-specific biomechanical issues,
 - 105
 - sport-specific hand conditions, 490
 - sport-specific rehabilitation programs, 6
 - sports-specific phase (adductor-related groin pain), 648-9
 - sprain
 - ligament, 14, 19-21
 - midtarsal joint sprains, 952
 - muscle, 14, 21-3
 - Strategic Assessment of Risk and Risk Tolerance (StARRT), 285-93
 - additional perspectives, 289
 - beyond risk for injury, 287
 - decision making, 290-3
 - outcomes and probabilities, 288-9
 - risk tolerance modifiers, 287
 - tissue health, 286-7
 - tissue stress, 287
 - static stretching, 149
 - sternoclavicular joint problems,
 - 517-18
 - strength testing and injuries, 215
 - strength training, 148
 - strengthening exercises (lateral elbow tendinopathy), 445-6
 - stress fractures, 34, 38, 460
 - anterior cortex of the tibia, 838
 - calcaneal, 947-8
 - cuboid, 956
 - cuneiforms, 956
 - diaphysis, 965
 - femur, 674-6
 - fibula, 845
 - great toe, 972
 - inferior pubic ramus, 656
 - medial malleolar, 928
 - navicular, 949-52
 - neck of metatarsals I-IV, 959-61
 - neck of the femur, 655-7
 - pars interarticularis, 535-43
 - and preventing overuse injuries, 182
 - rib stress fractures (RFS), 132, 515
 - special metatarsal stress fracture:
 - base of the second MT, 961
 - specific treatment of, 38
 - talus, 933
 - tibia, 836-8
 - stretching, 150
 - and preventing overuse injuries, 181-2
 - and recovery, 193-4, 200
 - structural ('static') biomechanical assessment, 96-100
 - ankle dorsiflexion, 98-9
 - foot, 96-8
 - leg length, 99-100
 - summary, 100
 - tibiofemoral alignment at the knee, 99
 - subacromial impingement, 381
 - subconjunctival haemorrhage, 338-9
 - subluxation, 14, 18-19
 - suboptimal lower limb biomechanics,
 - conditions related to, 106-8
 - subscapularis muscle tears, 429
 - subtler joint, 85-8
 - subungual exostosis, 972
 - subungual haematoma, 972
 - superficial compartment syndrome,
 - 863
 - superior tibiofibular joint injury, 767,
 - 815-16
 - supracondylar fractures, 456
 - suprascapular nerve
 - entrapment, 429-30
 - surfaces, injury prevention and sports, 187-8
 - surgery
 - Achilles tendon 884
 - arthroscopic 274-5, 621-2
 - hand injuries, 505
 - hip injury, 621-2
 - lateral elbow tendinopathy, 451
 - meniscal injuries, 727
 - open, 275
 - and patellofemoral pain, 789-90
 - and sports injuries, 274-5
 - wrist, 486
 - sustained myofascial tension, 254
 - swelling and injury, 212
 - 'swimmer's shoulder' *see* rotator cuff tendinopathy
 - swimming (biomechanical aspects of injuries), 133-6
 - biomechanics 133
 - freestyle techniques, 133
 - medial knee pain, 136
 - shoulder pain, 133-6
 - swimming biomechanics, 133
 - synovial plica, 803
 - synovitis 612-13
 - post-traumatic 913-14
- ## T
- T₄ syndrome, 514
 - talar dome, osteochondral lesions of the, 906-8
 - talus, stress fracture, 933
 - taping
 - finger injuries, 494
 - lateral elbow tendinopathy, 447-8
 - lower limb biomechanical issues, 112-14
 - musculoskeletal conditions, 259-61
 - tarsal coalition, 957-8, 988
 - tarsal navicular bone, apophysitis, 989
 - tarsal tunnel syndrome, 926-8
 - team care throughout the season,
 - 1018-20
 - anti-doping, 1020
 - core principles, 1018-20
 - teams, working and travelling with, 1017-25
 - emergency action plans, 1020-21
 - hygiene, 1020
 - immunisation and WHO, 1020
 - jet lag, symptomatic treatment for, 1025
 - jet lag and air travel, 1023
 - jet lag prevention, 1023
 - medical equipment, 1018
 - medical indemnity and trauma training, 1018
 - medical support team, 1017
 - medical team attributes, 1017-18
 - sexual activity, 1020
 - travel preparation, 1021-2
 - vaccinations, 1024
 - tear
 - fascia, 14, 26
 - ligament, 14, 19-21
 - muscle, 14, 21-5
 - tendon, 14, 25-6, 50
 - technique training, 143
 - teeth injuries, 340-3
 - dental management, 341
 - emergency management, 340-1
 - emergency treatment, 342
 - prevention, 341-3
 - temporomandibular injuries, 345-6
 - tendinopathy, 46-51
 - biceps femoris, 815, 823
 - degenerative, 49
 - extensor, 956
 - flexor hallucis longus, 924-6
 - gastrocnemius, 824
 - lower limb rehabilitation, 50-1

- midportion Achilles, 872-84
- patellar, 771, 793-801, 986
- peroneal, 929-31, 956
- pes anserinus, 819
- popliteus, 821
- proximal hamstring, 577-81
- quadriceps, 802
- tibialis anterior, 935-6
- tibialis posterior, 923-4, 955
- upper hamstring, 710-11
- tendinosis, 47
- tendon
 - dislocation/rupture (acute ankle injury), 911-12
 - dysrepair, 49
 - enthesopathy, 51
 - extensor carpi ulnaris tendon injuries, 484-5
 - hamstring ruptures, 767
 - hand and finger injuries, 499-504
 - and hyaluronic acid, 270
 - intramuscular tendon injuries, 690
 - and mechanotherapy, 245
 - and NSAID, 266
 - overuse injury (tendinopathy), 46-50, 143
 - partial tears, 50
 - peroneal dislocation, 911-12
 - rupture (patellar), 765
 - ruptures (acute elbow injuries), 459-60
 - tear/rupture, 14, 25-6, 50
 - tibialis posterior rupture, 912
 - tibialis posterior tendon dislocation, 912
- tennis (biomechanical aspects of injuries), 136-8
 - lateral elbow pain, 136-7
 - shoulder injuries, 137-8
- tennis elbow, 136-7, 440
- tenosynovitis
 - de Quervain's, 476-7
- thigh pain *see* anterior thigh pain; posterior thigh pain
- therapeutic exercise, 241-6
- therapeutic medication in musculoskeletal injury, 264-72
 - actovegin, 269
 - analgesics, 264
 - bisphosphonates, 272
 - central sensitisation, 267-8
 - corticosteroid, 270-71
 - GTN patches, 272
 - hyaluronic acid, 269-70
 - mechanical and high-volume injections, 269
 - neuropathic pain, 267-8
 - non-steroidal anti-inflammatory drugs (NSAIDs), 264-7
 - prolotherapy, 269
 - quinolone antibiotics, 272
 - sclerosant, 269
 - sleep medications, 271
 - traumeel, 269
- thoracic outlet syndrome (TOS), 430-1
- thoracic pain, 507-20
 - assessment, 508-11
 - causes, 507
 - costovertebral joint disorders, 511
 - intervertebral disc prolapse, 512-14
 - intervertebral joint disorders, 511
 - postural imbalance, 514
 - Scheuermann's disease, 512
 - T₄ syndrome, 514
- throwing
 - abnormal biomechanics, 118
 - acceleration, 115
 - biomechanics, 114-17
 - cocking, 114-15
 - deceleration/follow-through, 116
 - normal biomechanics of the scapular, 117-18
 - scapular biomechanics in shoulder injuries, 118-19
 - and shoulder injuries, 435-7
 - wind-up, 114
- tibia, stress fracture, 836-8, 846
 - anterior cortex, 838
 - assessment, 836-7
 - medial tibial stress syndrome, 839-41
 - prevention of recurrence, 837-8
 - treatment, 838
- tibial plafond lesions, 910
- tibialis anterior tendinopathy, 935-6
- tibialis posterior tendinopathy, 923-4, 955
- tibialis posterior tendon
 - dislocation, 911-12
- tibialis posterior tendon
 - rupture, 911-12
- tibiofemoral alignment at the knee, 99
- tibiofibular ligament injury/syndesmotic injury, anteroinferior, 912-13
- tissue healing
 - bone stress injury management, 42
- tissue status (damage), 56-7, 66
- toe clawing, 972
- training
 - agility, 144
 - and bone stress injuries, 32-3
 - Central Government Model (CGM) for the limits of performance, 140-2
 - cross-training, 143
 - endurance, 142-3
 - Fartlek, 142-3
 - flexibility, 149-50
 - hypertrophy, 149
 - individualisation, 140
 - interval, 142
 - load management for performance enhancement, 150-1
 - Maximal Aerobic Speed (MAS), 143
 - overload, 140
 - periodisation, 139-40
 - power, 148-9
 - principles of, 139-42
 - programming and prescription, 139-51
 - resistance, 144-9
 - specificity, 140
 - speed, 143-4
 - speed intervals, 143
 - strength, 148
 - technique, 143
 - see also* conditioning training; skill training
- training history and injury, 213
- training programs and preventing overuse injuries, 181-3
- transient receptor potential (TRP), 267-8
- transcutaneous electrical nerve stimulation (TENS), 262
- trapezium fracture, 475-6
- traumatic bursitis, 26
- traumatic shoulder instability (TUBS), 407-11
- traumeel, 269
- treatment
 - Achilles tendinopathy, 887
 - acute Achilles tendon rupture, 889-92
 - acute ankle injuries, 900-4, 907-8
 - adductor-related groin pain, 644-7
 - anterior impingement of the ankle, 935
 - anterolateral impingement, 932
 - chronic exertional compartment syndrome (CECS), 841-5
 - and concussion, 311
 - distal quadriceps muscle strain, 671-2
 - flexor hallucis longus tendinopathy, 926
 - gastrocnemius muscle strains, 853-6
 - hip-related pain, 614-22, 625-8
 - iliotibial band friction syndrome (IBTFS), 811-12
 - iliopsoas-related groin pain, 650-1
 - inguinal-related groin pain, 652-4
 - lateral elbow tendinopathy, 445
 - lateral ligament injuries, 899-900
 - lisfranc joint injuries, 954-5
 - medial calcaneal nerve entrapment, 929
 - medial collateral ligament injury, 729-36

medial malleolar stress fracture, 928
 medial tibial stress syndrome, 837
 meniscal injuries, 726-7
 midportion Achilles tendinopathy, 877-84
 options for pain, 72-4
 patellar dislocation, 765
 patellofemoral pain, 782-92
 peroneal tendinopathy, 931
 plantar fasciopathy, 943-6
 primary patellofemoral instability, 792-3
 pubic-related groin pain, 654
 quadriceps contusion, 666-770
 sacroiliac joint dysfunction, 585
 sinus tarsi syndrome, 932
 sport and exercise medicine (SEM), 5
 stress fracture of the
 tibia, 837-41
 tarsal tunnel syndrome, 927-8
 tibialis anterior tendinopathy, 935-6
 tibialis posterior tendinopathy, 924
 triangular fibrocartilage complex tear (TFCC), 483-4
 tricyclic antidepressants, 267
 trigger points
 lateral elbow tendinopathy, 447
 myofascial, 43-5, 68
 pain from, 44, 83
 posterior thigh, 707
 treatment, 44-5
 triquetral fracture, 482
 trunk function assessment, 102-5
 tumours
 bone, 78-80
 soft tissue, 78-80
 turf toe, 966

U

ulnar column presentations, 481-5
 ulnar nerve compression, 486
 ulnar styloid fracture, 481
 ultrasound, 37, 40
 acute knee injuries, 723
 for diagnosis, 228-30
 groin pain, 642
 and injection therapy, 268
 shoulder investigations, 401
 therapeutic, 261
 Ummukulthoum Bakare, 1031
 upper arm pain, 461-2
 upper limb 114-20
 abnormalities specific to
 throwing, 120
 changes in throwing arm with
 repeated throwing, 119-20
 biomechanics of the scapular in
 throwing, 117-18
 kinetic change, 116-17

neurodynamics test 215, 220, 258,
 354, 396
 shoulder injuries, 118-19
 throwing, 114-17

V

vacuum cupping, 255
 vascular disorders, 82
 vascular entrapment, 846
 vertebral endplate oedema, 547-8
 vitamin D
 bone stress injuries, 34
 musculoskeletal injuries, 273
 vitreous haemorrhage, 339

W

warm-up program, 178-81
 water immersion and recovery, 195-6
 West, Dr Liam, 1031-2
 whiplash associated disorders (WAD), 357, 375
 Whiteley, Rod, 1033
 work activities and injury, 212
 World Anti-Doping Agency (WADA), 1007
 wounds, treatment of open, 27
 wrist injuries and rowing, 132-3
 wrist pain, 463-87
 carpal bone dislocation, 479
 carpal tunnel syndrome, 471, 485-6
 central column problems, 478-81
 clinical approach, 463-70
 de Quervain's tenosynovitis, 476-7
 distal radius fracture, 470-2
 ganglions, 478-9
 impingement syndromes, 481
 intersection syndrome, 477-8
 investigations, 466-70
 Keinböck's disease, 479-81
 numbness, 485-6
 outcome measures, 466
 patient-reported outcome
 measures, 471
 physical examination, 466
 plain radiography, 466-8
 radial column problems, 470-8
 radial epiphyseal injury (gymnast's
 wrist), 476
 radial sensory nerve
 compression, 478
 regions, 464
 rehabilitation, 475, 487
 scapholunate dissociation, 479
 scaphoid fracture, 472-5
 special imaging studies, 468-70
 surgery, 486
 trapezium fracture, 475
 triangular fibrocartilage complex
 tear (TFCC), 483

ulnar column problems, 481-6
 ulnar nerve compression, 486
 young athletes, 980
 wrist pain and golf, 128-9
 leading wrist, 128-9
 trailing wrist, 129
 wrist splinting, 487
 wry neck, 374

X

X-ray, 401

Y

younger athlete, 974-89
 acute fractures, 976-9
 apophysitis, 982-3
 apophysitis of the fifth metatarsal,
 989
 apophysitis of the tarsal navicular
 bone, 989
 back pain, 981-2
 elbow pain, 980
 flat foot, 989
 foot pain, 988-9
 Freiberg's infarction, 989
 hip pain, 982-4
 idiopathic scoliosis, 981
 irritable hip, 984
 knee pain, 984-6
 Köhler's disease, 988-9
 maturity-associated variations, 975
 musculoskeletal conditions,
 management of, 976-89
 nonlinearity of growth, 974-5
 Osgood-Schlatter disease, 984-6
 overuse injuries of the physis, 979
 painless abnormalities of gait, 986-8
 patellar tendinopathy, 986
 Perthes' disease, 983
 popliteal cysts, 986
 postural abnormalities, 981-2
 referred pain from the hip, 986
 Sever's disease, 988
 shoulder pain, 979-80
 Sinding-Larsen-Johansson
 disease, 986
 slipped capital femoral
 epiphysis, 983-4
 spondylolisthesis, 981-2
 spondylolysis, 981
 tarsal conditions, 988
 unique response of skeletal injury,
 975-76
 uniqueness, 974-6
 wrist pain, 980-81

Z

zygomaticomaxillary complex, 344